

CS-728 Assignment 3

Deevyanshu Malu (22B1029) Deeptanshu Malu (22B1256)
Dhvanil Gheewala (22B0923) Tanmay Mandaliya (22B1037)

March 2026

1 Part 1: Classical Retrieval

In this part, We implemented standard retrieval methods where queries and tools are encoded independently, and similarity is used to retrieve relevant tools. Based on the similarity, the top-k most relevant tools were retrieved.

A table with recall@1 and recall@5 for three baselines is given below:

Retrieval methods	Recall@1	Recall@5
BM25	0.186	0.3476
msmarco-MiniLM	0.3514	0.5716
UAE-large-v1	0.6142	0.865

Table 1: recall@1 and recall@5 for the three baselines

2 Part 2: Lost-in-the-Middle

In this part, we move to a different setting where the query and all candidate tools are placed together in a single prompt and processed jointly by a language model. The goal is to use attention inside the model as a retrieval signal, and the correlation between this signal and the position of the relevant tool.

2.1 Table with recall@1 and recall@5

Similar to Part 1, we report a table containing recall@1 and recall@5:

Recall@1	Recall@5
0.0122	0.2718

Table 2: recall@1 and recall@5 values

2.2 Plot to visualize position effects on attention based ranking

To visualise the correlation between this signal and the position of relevant tool, we attach the following plot:

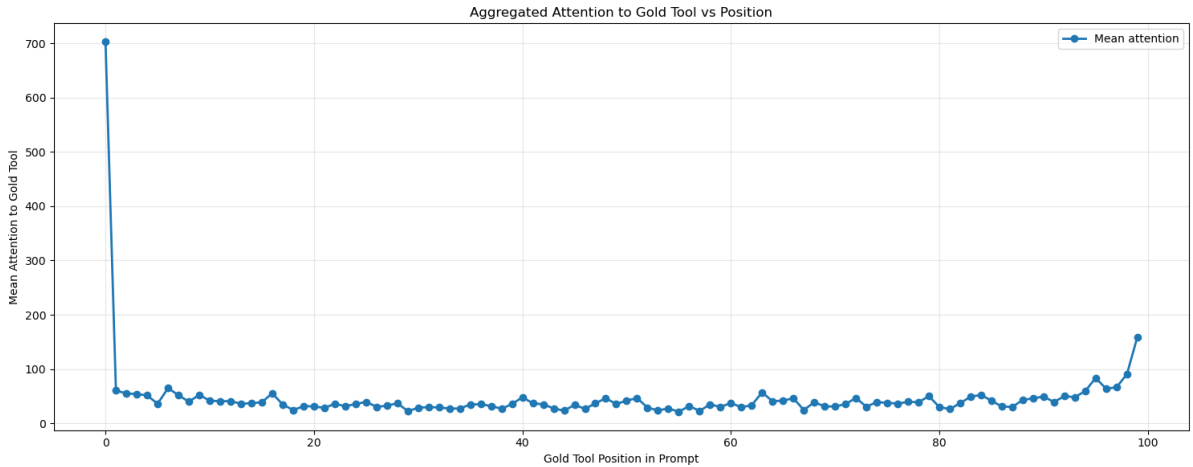


Figure 1: gold tool attention vs tool position

3 Part 3: Retrieval Heads

In the previous part, we used attention aggregated across all heads to score tools. In this part, we take a small subset of attention heads that act as effective retrievers. We first try to select useful heads using training data and then use only those heads for validation.

3.1 Attention Head Selection

(BONUS) We use 2 methods to calculate attention scores.

Normalized Attention Score. This metric measures how much attention a model head allocates to the correct (gold) tool relative to all tools, computed per query and then aggregated over a set of queries.

Let $\mathcal{Q}$ denote a set of queries. For each query $q \in \mathcal{Q}$, let Q_q denote its set of query tokens, and let T_t denote the set of tokens corresponding to tool t . Let $A_{l,h}(i, k)$ be the attention weight from token i to token k at layer l and head h , and let t^* denote the gold tool.

For each query q , define the tool-wise attention:

$$s_{t,l,h}^{(q)} = \sum_{i \in Q_q} \sum_{k \in T_t} A_{l,h}(i, k)$$

The per-query normalized score is:

$$\text{Score}_{l,h}^{(q)} = \frac{s_{t^*,l,h}^{(q)}}{\sum_t s_{t,l,h}^{(q)} + \epsilon}$$

The final score is obtained by summing over all queries:

$$\text{Score}_{l,h} = \sum_{q \in \mathcal{Q}} \text{Score}_{l,h}^{(q)}$$

Rank-based Attention Score. This metric evaluates how highly the correct (gold) tool is ranked among all tools based on attention, computed per query and then aggregated over a set of queries.

Let $\mathcal{Q}$ denote a set of queries. For each query $q \in \mathcal{Q}$, let Q_q denote its set of query tokens, and let T_t denote the set of tokens corresponding to tool t . Let $A_{l,h}(i, k)$ be the attention weight from token i to token k at layer l and head h , and let t^* denote the gold tool.

For each query q , define:

$$s_{t,l,h}^{(q)} = \sum_{i \in Q_q} \sum_{k \in T_t} A_{l,h}(i, k)$$

Let $\text{rank}_{l,h}^{(q)}(t^*)$ denote the rank of t^* when tools are sorted in descending order of $s_{t,l,h}^{(q)}$.

The per-query rank-based score is:

$$\text{Score}_{l,h}^{(q)} = \frac{1}{\text{rank}_{l,h}^{(q)}(t^*) + 1}$$

The final score is:

$$\text{Score}_{l,h} = \sum_{q \in \mathcal{Q}} \text{Score}_{l,h}^{(q)}$$

3.2 A table with recall@1 and recall@5

(BONUS) The following tables display the recall performances when tested with 10, 20, and 30 as the maximum number of heads selected:

max_heads=10

aggregation methods	Recall@1	Recall@5
normalised	0.3742	0.6314
rank	0.4006	0.6856

Table 3: recall@1 and recall@5 when max_heads=10

The heads selected when normalised attention values were used were:

Layer	Head
6	11
5	11
10	1
6	8
8	19
10	23
6	9
8	13
10	9
2	20

Table 4: Selected top attention heads using rank-based aggregation (layer, head).

The heads selected when rank-based attention values were used were:

Layer	Head
6	11
7	12
8	19
2	20
8	13
8	20
4	16
6	8
5	11
10	9

Table 5: Selected top attention heads (layer, head).

max_heads=20

aggregation methods	Recall@1	Recall@5
normalised	0.285	0.62
rank	0.3718	0.666

Table 6: recall@1 and recall@5 when max_heads=20

The heads selected when normalised attention values were used were:

Layer	Head
6	11
5	11
10	1
6	8
8	19
10	23
6	9
8	13
10	9
2	20
8	20
12	13
5	10
7	12
10	20
12	15
10	13
10	26
10	15
13	20

Table 7: Selected top 20 attention heads using normalised attention values (layer, head).

The heads selected when rank-based attention values were used were:

Layer	Head
6	11
7	12
8	19
2	20
8	13
8	20
4	16
6	8
5	11
10	9
10	1
10	23
6	9
5	9
8	17
5	10
10	11
8	21
10	20
5	5

Table 8: Selected top 20 attention heads using rank-based aggregation (layer, head).

max_heads=30

aggregation methods	Recall@1	Recall@5
normalised	0.2492	0.5936
rank	0.1638	0.628

Table 9: recall@1 and recall@5 when max_heads=30

The heads selected when normalised attention values were used were:

Layer	Head
6	11
5	11
10	1
6	8
8	19
10	23
6	9
8	13
10	9
2	20
8	20
12	13
5	10
7	12
10	20
12	15
10	13
10	26
10	15
13	20
8	17
4	16
10	10
12	2
11	25
11	20
7	5
11	22
11	27
8	21

Table 10: Selected top 30 attention heads using normalised attention values (layer, head).

The heads selected when rank-based attention values were used were:

Layer	Head
6	11
7	12
8	19
2	20
8	13
8	20
4	16
6	8
5	11
10	9
10	1
10	23
6	9
5	9
8	17
5	10
10	11
8	21
10	20
5	5
10	26
10	13
12	13
13	20
10	15
8	30
8	24
12	15
1	23
5	8

Table 11: Selected top 30 attention heads using rank-based aggregation (layer, head).

3.3 Comparing the recall performances of different parts

As demonstrated in Parts 2 and 3, recall performance drops with the increase in the number of heads selected. In our experiments, this optimal number of heads is 10. As mentioned above, we observe a drop in performance as the number of heads increases from 10 to 30, and in Part 2, where all available heads are utilized. In Part 1, recall performance using BM25 and `msmarco-MiniLM` retrieval methods is worse than when 10-30 heads are selected, better than when all heads are selected. However, retrieval performance using the `UAE-large-v1` model is very strong and the best among all methods.

4 GenAI Usage

<https://chatgpt.com/share/69cbb2cf-b7fc-83a6-8633-1de096281048>